

5(a)(i) Max induced e.m.f. occurs when gradient of graph is steepest, which is from 3.0 to 4.0 s.

From Faraday's law, magnitude $\varepsilon = \left| -\frac{d\phi}{dt} \right|$

$$= \left| -\frac{d(NBA)}{dt} \right|$$

$$= (500)(\pi)(0.12)^2 \left[\frac{0.0024}{4-3} \right]$$

$$= 0.054 \text{ V}$$

(a)(ii)

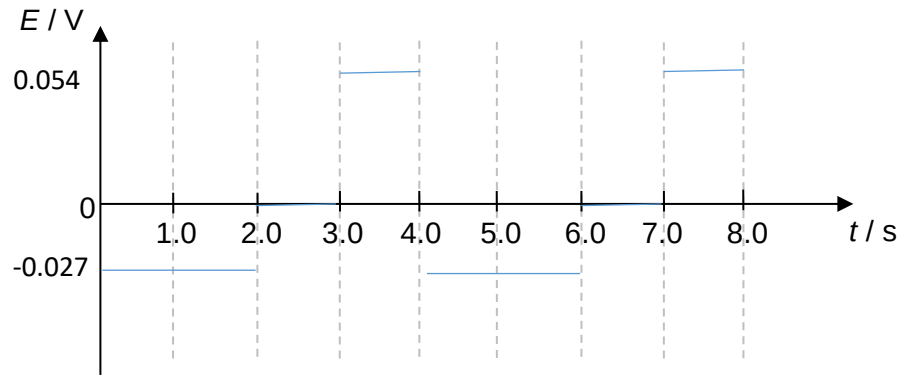


Fig. 5.3

(b)(i) For a coil with N turns, $B = \frac{\mu_o NI}{2r}$

From Faraday's law,

magnitude of induced emf in ring $\varepsilon = \left| -\frac{d\phi}{dt} \right|$

$$= \left| -\frac{d(BA)}{dt} \right|$$

$$= \frac{d\left(\frac{\mu_o N A I_o \sin \omega t}{2r}\right)}{dt} = \frac{\mu_o N A}{2r} \frac{d(I_o \sin \omega t)}{dt}$$

$$= \frac{\mu_o N A}{2r} I_o \omega \cos \omega t$$

Max when $\cos \omega t = 1$

$$\text{Maximum induced emf, } E_o = (0.025) \left[\frac{(4\pi \times 10^{-7})(500)}{2(0.12)} \right] I_o \omega$$

$$= (6.54498 \times 10^{-5}) I_o \omega$$

$$= (6.54 \times 10^{-5}) I_o \omega$$

(b)(ii) $\omega = 2\pi f = 2\pi(50) = 100\pi \text{ rad s}^{-1}$

Peak current in coil, $I_o = \sqrt{2} I_{rms} = \sqrt{2}(3.5) = 4.95 \text{ A}$

$$E_o = (6.64 \times 10^{-5})(4.95)(100\pi) = 0.102 \text{ V}$$

$$\text{Maximum induced current, } I = \frac{E_o}{R} = \frac{0.102}{0.2} = 0.51 \text{ A}$$

(b)(iii) Induced current remains the same since both induced emf and resistance will be doubled.

- (c)** Alternating current flowing in the coil produces a changing magnetic flux linkage in the pot.
According to Faraday's law, this causes an induced emf and hence current in the pot.
As the induced current flows through the resistance of the pot, heat is generated. This causes the pot to heat up.